

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 18 — Carboxylic Acids and Derivatives (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	18.1 Carboxylic Acids: Structure, Acidity, Inductive Effects & Reactions · 18.2 Esters: Formation, IUPAC Nomenclature
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

- | | |
|--|--|
| 18.1 Carboxylic Acids: Structure, Acidity & Reactions | Cyclic hydrogen-bonded dimers, high boiling points; weak acidity (K_a/pK_a) and resonance delocalization in the carboxylate group. |
| 18.2 Esters: Formation, Naming & Hydrolysis | — Esterification of carboxylic acids with alcohols catalyzed by concentrated H_2SO_4 ; rapid irreversible ester synthesis. |
| High-Frequency Core Repeats | — The 10 most frequently examined Cambridge Paper 1 questions on Carboxylic Acids and Esters (Q101 – Q110). |

SUBTOPIC 18.1 — CARBOXYLIC ACIDS: STRUCTURE, ACIDITY & REACTIONS (Q1 – Q55)

1 [EASY] Dimerisation and High Boiling Point of Carboxylic Acids — 9701/11/M/J/23/Q29

[1]

Ethanoic acid ($M_r = 60.0$) boils at 118°C , whereas propan-1-ol ($M_r = 60.0$) boils at 97°C .

What accounts for the significantly higher boiling point of ethanoic acid?

- [A] Ethanoic acid has a much larger electron cloud than propan-1-ol.
- [B] Carboxylic acid molecules form two reciprocal intermolecular hydrogen bonds between their $\text{C}=\text{O}$ and $\text{O}-\text{H}$ groups, existing as stable cyclic dimers in the liquid state.
- [C] Ethanoic acid is completely ionic in the pure liquid state.
- [D] Propan-1-ol cannot form intermolecular hydrogen bonds.

2 [HARD] Stability of the Carboxylate Anion and Acidity — 9701/12/M/J/23/Q29

[1]

Why are carboxylic acids significantly stronger Bronsted-Lowry acids ($\text{pK}_a \sim 4-5$) than aliphatic alcohols ($\text{pK}_a \sim 16$)?

- [A] The $\text{O}-\text{H}$ bond in carboxylic acids is completely ionic.
- [B] Carboxylic acids dissociate completely in aqueous solution.
- [C] The alkyl group donates electron density to stabilize the proton.
- [D] Deprotonation yields a carboxylate anion (RCOO^-) in which the negative charge is delocalised equally over both electronegative oxygen atoms via a delocalised π system, forming two identical $\text{C}-\text{O}$ bonds.

3 [HARD] Inductive Effects of Chlorine Substituents on Acid Strength — 9701/13/M/J/23/Q29

[1]

Which sequence correctly ranks the following acids in order of INCREASING acid strength (lowest K_a to highest K_a)?

- [A] Ethanoic acid < Chloroethanoic acid < Dichloroethanoic acid < Trichloroethanoic acid
- [B] Trichloroethanoic acid < Dichloroethanoic acid < Chloroethanoic acid < Ethanoic acid
- [C] Ethanoic acid < Trichloroethanoic acid < Dichloroethanoic acid < Chloroethanoic acid
- [D] Chloroethanoic acid < Ethanoic acid < Dichloroethanoic acid < Trichloroethanoic acid

4 [EASY] Distinguishing Carboxylic Acids from Alcohols and Phenols — 9701/11/O/N/23/Q29

[1]

Which chemical reagent reacts with ethanoic acid to produce vigorous effervescence of a gas that turns limewater milky, but produces NO reaction with ethanol or phenol?

- [A] Sodium metal
- [B] Acidified potassium dichromate(VI)
- [C] Aqueous sodium hydrogencarbonate (NaHCO_3) or sodium carbonate (Na_2CO_3)
- [D] Phosphorus(V) chloride

5 [EASY] Position of Halogen Substituent and Acid Strength — 9701/13/M/J/220/Q5

[1]

How does the position of the chlorine atom affect the acid strength of chlorobutanoic acid isomers?

- [A] 2-chlorobutanoic acid is the strongest acid because the electronegative chlorine atom is closest to the carboxylate group, maximizing the $-I$ charge-dispersal effect.
- [B] 4-chlorobutanoic acid is the strongest acid because chlorine is further away.
- [C] All chlorobutanoic acid isomers have identical pK_a values.
- [D] 3-chlorobutanoic acid is the strongest due to steric hindrance.

6 [HARD] Effect of Alkyl Group on Acidity (Methanoic vs Ethanoic) — 9701/11/M/J/221/Q6

[1]

Why is methanoic acid (HCOOH , $\text{pK}_a = 3.75$) a stronger acid than ethanoic acid (CH_3COOH , $\text{pK}_a = 4.76$)?

- [A] Methanoic acid contains two oxygen atoms while ethanoic acid has only one.
- [B] Ethanoic acid forms stronger hydrogen bonds in water.
- [C] Methanoic acid is an inorganic mineral acid.
- [D] The methyl group in ethanoic acid exerts an electron-donating (+I) inductive effect, which pushes electron density onto the carboxylate group, intensifying the negative charge and destabilizing the anion.

7 [EASY] Fluoroethanoic vs Chloroethanoic vs Bromoethanoic Acid — 9701/12/M/J/222/Q7

[1]

Which sequence ranks the haloethanoic acids in order of DECREASING acid strength (strongest to weakest)?

- [A] Iodoethanoic acid > Bromoethanoic acid > Chloroethanoic acid > Fluoroethanoic acid
- [B] Fluoroethanoic acid > Chloroethanoic acid > Bromoethanoic acid > Iodoethanoic acid
- [C] Chloroethanoic acid > Fluoroethanoic acid > Bromoethanoic acid > Iodoethanoic acid
- [D] All four haloethanoic acids have identical K_a values.

8 [HARD] Reaction of Carboxylic Acids with Reactive Metals — 9701/13/M/J/223/Q8

[1]

What is observed when magnesium ribbon is added to a test tube of dilute propanoic acid?

- [A] A violent explosion with oxygen evolution.
- [B] Formation of a dark brown precipitate of copper.
- [C] Slow effervescence of hydrogen gas (tested with a squeaky pop) and magnesium dissolves to form magnesium propanoate:
 $2\text{CH}_3\text{CH}_2\text{COOH} + \text{Mg} \rightarrow (\text{CH}_3\text{CH}_2\text{COO})_2\text{Mg} + \text{H}_2(\text{g})$.
- [D] No reaction occurs because propanoic acid is weak.

9 [EASY] Reaction of Carboxylic Acids with Metal Oxides — 9701/11/M/J/224/Q9

[1]

What are the products when ethanoic acid is warmed with black copper(II) oxide powder?

- [A] Copper metal, carbon dioxide, and steam.
- [B] A blue-green solution of copper(II) ethanoate, $(\text{CH}_3\text{COO})_2\text{Cu}$, and water.
- [C] Copper(I) oxide and ethanal.
- [D] A bright yellow precipitate of sulfur.

10 [HARD] Reaction of Carboxylic Acids with Alkalis — 9701/12/M/J/220/Q10

[1]

What is the standard enthalpy of neutralization for the reaction between dilute ethanoic acid and aqueous sodium hydroxide, and why is it less exothermic than that for $\text{HCl} + \text{NaOH}$ ($-57.1 \text{ kJ mol}^{-1}$)?

- [A] $-55.2 \text{ kJ mol}^{-1}$; some energy is consumed in ionizing the weak, partially dissociated ethanoic acid molecules before neutralization.
- [B] $-57.1 \text{ kJ mol}^{-1}$; all neutralizations have identical enthalpy.
- [C] -100 kJ mol^{-1} ; ethanoic acid releases extra heat.
- [D] Endothermic ($+25 \text{ kJ mol}^{-1}$).

11 [EASY] Reduction of Carboxylic Acids with LiAlH_4 — 9701/13/M/J/221/Q11

[1]

What is the product when propanoic acid is treated with lithium aluminium hydride (LiAlH_4) in dry ether followed by dilute acid?

- [A] Propanal
- [B] Propan-2-ol
- [C] Propane
- [D] Propan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (a primary alcohol)

12 [HARD] Resistance of Carboxylic Acids to NaBH_4 — 9701/11/M/J/222/Q12

[1]

Why does sodium borohydride (NaBH_4) FAIL to reduce carboxylic acids to alcohols?

- [A] NaBH_4 is an oxidizing agent.
- [B] Carboxylic acids are completely insoluble in all solvents.
- [C] Carboxylic acids are rapidly deprotonated by NaBH_4 to carboxylate anions (RCOO^-), which possess a negative charge that electrostatically repels the nucleophilic $[\text{BH}_4]^-$ / hydride ion.
- [D] NaBH_4 only reacts with alkanes.

13 [EASY] Preparation of Acyl Chlorides using PCl_5 — 9701/12/M/J/223/Q13

[1]

What is observed when solid phosphorus(V) chloride, PCl_5 , is added to glacial ethanoic acid at room temperature?

- [A] A vigorous exothermic reaction with the evolution of steamy acidic fumes of HCl gas, forming ethanoyl chloride (CH_3COCl).
- [B] A bright yellow precipitate of silver chloride.
- [C] The solution turns green.
- [D] Carbon dioxide gas is evolved.

14 [HARD] Preparation of Acyl Chlorides using SOCl_2 — 9701/13/M/J/224/Q14

[1]

Why is thionyl chloride, SOCl_2 , the preferred laboratory reagent for preparing acyl chlorides from carboxylic acids?

- [A] SOCl_2 is completely non-toxic and non-corrosive.
- [B] SOCl_2 is solid at room temperature.
- [C] Both byproducts (SO_2 and HCl) are gases that escape from the reaction mixture, leaving pure acyl chloride without requiring complex distillation.
- [D] It produces water as the only byproduct.

15 [EASY] Preparation of Acyl Chlorides using PCl_3 — 9701/11/M/J/220/Q15

[1]

What are the products when three moles of ethanoic acid react with one mole of phosphorus(III) chloride, PCl_3 ?

- [A] Three moles of ethanoyl chloride, phosphorus, and chlorine gas.
- [B] Three moles of ethanoyl chloride (CH_3COCl) and one mole of phosphorous acid (H_3PO_3).
- [C] Ethanol and phosphoric acid.
- [D] Chloroethane and POCl_3 .

16 [HARD] Water Solubility of Carboxylic Acids — 9701/12/M/J/221/Q16

[1]

Why are methanoic, ethanoic, propanoic, and butanoic acids completely miscible with water, whereas hexanoic acid is only sparingly soluble?

- [A] Hexanoic acid reacts with water to form an insoluble ester.
- [B] Lower acids are gases at room temperature.
- [C] Hexanoic acid contains no oxygen atoms.
- [D] The polar carboxyl group forms strong hydrogen bonds with water molecules, but as the hydrophobic hydrocarbon chain lengthens, water-water attractions are disrupted without sufficient compensation.

17 [EASY] Bond Lengths in Carboxylic Acids vs Carboxylate Anions — 9701/13/M/J/222/Q17

[1]

In propanoic acid, there is a short $\text{C}=\text{O}$ bond (0.120 nm) and a longer $\text{C}-\text{O}$ bond (0.134 nm). In sodium propanoate, both carbon-oxygen bonds are exactly 0.127 nm. What explains this?

- [A] The sodium cation pulls the oxygen atoms closer.
- [B] Propanoic acid is an equilibrium mixture of two isomers.
- [C] In the carboxylate anion, the unshared p orbital electrons on oxygen are delocalised symmetrically across both oxygen atoms, making both $\text{C}-\text{O}$ bonds identical with a bond order of 1.5.
- [D] The $\text{C}-\text{C}$ bond expands to shorten the $\text{C}-\text{O}$ bonds.

18 [HARD] Oxidation of Methanoic Acid by Tollens' Reagent — 9701/11/M/J/223/Q18

[1]

Why does methanoic acid (HCOOH) reduce Tollens' reagent to form a silver mirror, while ethanoic acid (CH_3COOH) does NOT?

- [A] Methanoic acid contains a formyl / aldehyde-like grouping ($\text{H}-\text{C}=\text{O}$) attached to $-\text{OH}$, allowing it to undergo further oxidation to carbon dioxide and water.
- [B] Methanoic acid is an inorganic acid.
- [C] Ethanoic acid decomposes Tollens' reagent.
- [D] Methanoic acid is a stronger base than ammonia.

19 [EASY] Oxidation of Ethanedioic Acid (Oxalic Acid) — 9701/12/M/J/224/Q19**[1]**

What products and visual observation occur when ethanedioic acid, HOOC-COOH , is warmed with acidified potassium manganate(VII)?

- [A] The solution turns vivid green.
- [B] A brick-red precipitate is formed.
- [C] No reaction occurs.
- [D] Vigorous effervescence of carbon dioxide (CO_2) gas, and the purple MnO_4^- solution is decolorized to colorless Mn^{2+} .

20 [HARD] Preparation of Carboxylic Acids via Nitrile Hydrolysis — 9701/13/M/J/220/Q20**[1]**

What reagents and conditions are required to synthesize butanoic acid from 1-bromopropane in two steps?

- [A] Step 1: NaOH(aq) , reflux; Step 2: acidified potassium dichromate.
- [B] Step 1: KCN in ethanol, heated under reflux (forming butanenitrile); Step 2: Dilute hydrochloric acid, heated under reflux (hydrolysis).
- [C] Step 1: Mg in ether; Step 2: water.
- [D] Step 1: NH_3 in ethanol; Step 2: HNO_3 .

21 [EASY] Preparation of Carboxylic Acids via Grignard Carbonation — 9701/11/M/J/221/Q21**[1]**

What sequence converts bromobenzene ($\text{C}_6\text{H}_5\text{Br}$) into benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$)?

- [A] Heat with concentrated nitric acid.
- [B] React with Mg in dry ether to form phenylmagnesium bromide ($\text{C}_6\text{H}_5\text{MgBr}$), pour onto solid carbon dioxide (CO_2 / dry ice), then hydrolyze with dilute acid.
- [C] React with potassium cyanide in ethanol.
- [D] Boil with aqueous sodium hydroxide.

22 [HARD] Acidity: Benzoic Acid vs Ethanoic Acid — 9701/12/M/J/222/Q22**[1]**

How does the acid strength of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$, $\text{pK}_a = 4.20$) compare with ethanoic acid (CH_3COOH , $\text{pK}_a = 4.76$)?

- [A] Benzoic acid is weaker because benzene donates electrons.
- [B] Benzoic acid is completely neutral.
- [C] Ethanoic acid is stronger because it has more hydrogen atoms.
- [D] Benzoic acid is slightly stronger than ethanoic acid because the sp^2 hybridised phenyl ring carbon is more electronegative than the sp^3 methyl carbon, exerting a weak electron-withdrawing ($-\text{I}$) effect.

23 [EASY] Acidity: 4-Nitrobenzoic Acid vs Benzoic Acid — 9701/13/M/J/223/Q23**[1]**

Why is 4-nitrobenzoic acid ($\text{pK}_a = 3.44$) significantly stronger than benzoic acid ($\text{pK}_a = 4.20$)?

- [A] The strongly electron-withdrawing nitro group ($-\text{NO}_2$) withdraws electron density by both $-\text{I}$ inductive and $-\text{M}$ resonance effects, dispersing the negative charge on the carboxylate group.
- [B] The nitro group donates protons directly.
- [C] The nitro group forms hydrogen bonds with water.
- [D] Benzoic acid decomposes in the presence of nitro groups.

24 [HARD] Acidity: 4-Methoxybenzoic Acid vs Benzoic Acid — 9701/11/M/J/224/Q24**[1]**

Why is 4-methoxybenzoic acid ($\text{pK}_a = 4.47$) a WEAKER acid than benzoic acid ($\text{pK}_a = 4.20$)?

- [A] Methoxy is an electron-withdrawing group.
- [B] Methoxybenzoic acid is insoluble in water.
- [C] The lone pair on the methoxy oxygen delocalises into the aromatic ring ($+\text{M}$ resonance effect), increasing electron density on the carboxylate group and destabilizing the anion.
- [D] The methoxy group absorbs protons.

25 [EASY] Decarboxylation of Carboxylic Acids — 9701/12/M/J/220/Q25**[1]**

What reaction occurs when solid anhydrous sodium ethanoate is heated strongly with soda lime (NaOH / CaO)?

- [A] Decarboxylation to produce methane gas (CH_4) and sodium carbonate: $\text{CH}_3\text{COONa} + \text{NaOH} \rightarrow \text{CH}_4(\text{g}) + \text{Na}_2\text{CO}_3$.
- [B] Combustion to carbon dioxide and water.
- [C] Polymerisation into polyethene.
- [D] Reduction to ethanol.

26 [HARD] Dicarboxylic Acids: Thermal Dehydration — 9701/13/M/J/221/Q26**[1]**

What product is formed when butanedioic acid (succinic acid) is heated strongly?

- [A] Butane and carbon dioxide.
- [B] Ethene and oxalic acid.
- [C] 2-methylpropanoic acid.
- [D] Succinic anhydride (a cyclic 5-membered anhydride) and water.

27 [EASY] Oxidation of Primary Alcohols: Complete Conversion — 9701/11/M/J/222/Q27**[1]**

Why is an excess of acidified potassium dichromate(VI) and prolonged heating under reflux necessary to convert ethanol completely into ethanoic acid?

- [A] Ethanal boils at 150°C .
- [B] Reflux ensures that volatile intermediate ethanal does not escape, allowing repeated contact with the hot oxidizer until fully converted to ethanoic acid.
- [C] Refluxing removes water.
- [D] Dichromate only oxidizes ethanal when cold.

28 [HARD] Identification of Carboxylic Acid by Titration — 9701/12/M/J/223/Q28**[1]**

A 0.150 g sample of an unknown monoprotic carboxylic acid requires 25.0 cm^3 of $0.100\text{ mol dm}^{-3}\text{ NaOH}$ for complete neutralization. What is the acid?

- [A] Methanoic acid, HCOOH ($M_r = 46.0$)
- [B] Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$ ($M_r = 74.0$)
- [C] Ethanoic acid, CH_3COOH ($M_r = 60.0$)
- [D] Butanoic acid, $\text{C}_3\text{H}_7\text{COOH}$ ($M_r = 88.0$)

29 [EASY] Acyl Chloride Reaction with Water — 9701/13/M/J/224/Q29**[1]**

What is observed when ethanoyl chloride, CH_3COCl , is added dropwise to cold water?

- [A] Formation of a sweet fruity smell of an ester.
- [B] A violent exothermic reaction with steamy, choking acidic fumes of HCl gas, forming ethanoic acid.
- [C] A white precipitate of silver chloride.
- [D] No reaction occurs.

30 [HARD] Acyl Chloride Reaction with Alcohols — 9701/11/M/J/220/Q30**[1]**

What is the product when ethanoyl chloride reacts with propan-1-ol at room temperature?

- [A] Propyl ethanoate (an ester) and hydrogen chloride gas (HCl).
- [B] Ethyl propanoate and water.
- [C] Propanal and ethanoic acid.
- [D] Propanoic acid and chloroethane.

31 [EASY] Acyl Chloride Reaction with Ammonia — 9701/12/M/J/221/Q31

[1]

What products are formed when ethanoyl chloride is treated with concentrated aqueous ammonia?

- [A] Ethylamine and water.
- [B] Ethanenitrile and HCl.
- [C] Aminoethanoic acid.
- [D] Ethanamide (CH_3CONH_2 , a primary amide) and dense white fumes of ammonium chloride (NH_4Cl).

32 [HARD] Acyl Chloride Reaction with Primary Amines — 9701/13/M/J/222/Q32

[1]

What organic product is formed when ethanoyl chloride reacts with methylamine (CH_3NH_2)?

- [A] Ethanamide and chloromethane
- [B] Dimethylamine
- [C] N-methylethanamide, $\text{CH}_3\text{CONHCH}_3$ (a secondary amide)
- [D] Ethanenitrile

33 [EASY] Reactivity of Carboxylic Acid Derivatives Order — 9701/11/M/J/223/Q33

[1]

What is the correct sequence of reactivity of carboxylic acid derivatives towards nucleophilic attack (most reactive to least reactive)?

- [A] Acyl chloride > Acid anhydride > Ester > Amide
- [B] Amide > Ester > Acid anhydride > Acyl chloride
- [C] Ester > Acyl chloride > Amide > Acid anhydride
- [D] Acyl chloride > Amide > Ester > Acid anhydride

34 [HARD] Reaction of Carboxylic Acids with Diazomethane — 9701/12/M/J/224/Q34

[1]

What reaction converts carboxylic acids quantitatively into methyl esters at 0°C without acid catalyst?

- [A] Reaction with methane gas.
- [B] Reaction with methylamine.
- [C] Reaction with diazomethane (CH_2N_2) in ether, releasing nitrogen gas: $\text{RCOOH} + \text{CH}_2\text{N}_2 \rightarrow \text{RCOOCH}_3 + \text{N}_2(\text{g})$.
- [D] Reaction with methanol and sodium.

35 [EASY] Natural Occurrence of Carboxylic Acids — 9701/13/M/J/220/Q35

[1]

Which pair correctly matches the carboxylic acid with its natural source?

- [A] Ethanoic acid in sour milk.
- [B] Methanoic acid in ant stings and stinging nettles; Ethanoic acid in vinegar; Citric acid in citrus fruits.
- [C] Methanoic acid in rancid butter.
- [D] Propanoic acid in lemons.

36 [HARD] Biological Decarboxylation — 9701/11/M/J/221/Q36

[1]

What gas is produced in cellular respiration when pyruvate ($\text{CH}_3\text{COCOO}^-$) undergoes oxidative decarboxylation to acetyl-CoA?

- [A] Carbon monoxide, CO
- [B] Oxygen, O_2
- [C] Methane, CH_4
- [D] Carbon dioxide, CO_2

37 [EASY] Physical State of Carboxylic Acids — 9701/12/M/J/222/Q37

[1]

What is the physical state of methanoic, ethanoic, and octadecanoic (stearic) acids at room temperature (20°C)?

- [A] All three are gases.
- [B] All three are solids.
- [C] Methanoic and ethanoic acids are colorless liquids; stearic acid is a waxy white solid.
- [D] Methanoic is a gas; others are liquids.

38 [HARD] Glacial Acetic Acid Name Origin — 9701/13/M/J/223/Q38**[1]**

Why is pure (100%) ethanoic acid commonly referred to as 'glacial' acetic acid in laboratories?

- [A]** It freezes into ice-like, colorless crystalline sheets at 16.6 °C (slightly below room temperature).
- [B]** It is mined from polar glaciers.
- [C]** It feels cold to the touch due to endothermic evaporation.
- [D]** It reacts violently with ice.

39 [EASY] Acidity of Carboxylic Acids in Non-Aqueous Solvents — 9701/11/M/J/224/Q39**[1]**

Why does ethanoic acid show negligible electrical conductivity when dissolved in pure benzene, but conducts electricity well when dissolved in water?

- [A]** Benzene destroys all covalent bonds.
- [B]** Ethanoic acid evaporates completely in benzene.
- [C]** Water neutralizes ethanoic acid.
- [D]** In non-polar benzene, ethanoic acid exists as un-ionized hydrogen-bonded dimers; in water, it ionizes into hydrated H_3O^+ and CH_3COO^- ions.

40 [HARD] Formation of Acid Anhydrides from Acyl Chlorides — 9701/12/M/J/220/Q40**[1]**

What compound is produced when ethanoyl chloride reacts with sodium ethanoate?

- [A]** Ethyl ethanoate
- [B]** Ethanoic anhydride, $(\text{CH}_3\text{CO})_2\text{O}$, and sodium chloride
- [C]** Ethanal
- [D]** Diethyl ether

41 [EASY] Reaction of Acyl Chlorides with Phenol — 9701/13/M/J/221/Q41**[1]**

Why do acyl chlorides react readily with phenol in the presence of base (Schotten-Baumann reaction), whereas carboxylic acids fail to esterify phenol?

- [A]** Phenol is an aliphatic alcohol.
- [B]** Acyl chlorides are far more electrophilic than carboxylic acids and react irreversibly with the weakly nucleophilic phenoxide ion.
- [C]** Carboxylic acids destroy phenol.
- [D]** Acyl chlorides react only with aromatic rings.

42 [HARD] Infrared Spectrum of Carboxylic Acids — 9701/11/M/J/222/Q42**[1]**

What two distinctive absorption bands characterize the infrared spectrum of a carboxylic acid?

- [A]** A sharp O-H peak at 3600 cm^{-1} and a C-C peak at 1200 cm^{-1} .
- [B]** A C-H stretch only at 2950 cm^{-1} .
- [C]** No absorption above 1500 cm^{-1} .
- [D]** A very broad, intense O-H stretch between $2500\text{--}3300\text{ cm}^{-1}$ centered at $\sim 3000\text{ cm}^{-1}$, and a sharp C=O stretch at $1700\text{--}1725\text{ cm}^{-1}$.

43 [EASY] Titration Curve of Weak Acid with Strong Base — 9701/12/M/J/223/Q43**[1]**

In the titration curve of ethanoic acid with sodium hydroxide, what is the pH at the equivalence point?

- [A]** $\text{pH} > 7$ (approximately pH 8.7) due to alkaline hydrolysis of the ethanoate anion ($\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$).
- [B]** $\text{pH} = 7.0$ exactly.
- [C]** $\text{pH} < 7$ (approximately pH 4.5).
- [D]** $\text{pH} = 14.0$.

44 [HARD] Buffer Solution Formation with Carboxylic Acids — 9701/13/M/J/224/Q44**[1]**

What chemical mixture constitutes an acidic buffer solution using ethanoic acid?

- [A] A mixture of hydrochloric acid and sodium chloride.
- [B] Pure ethanoic acid and water.
- [C] A mixture of ethanoic acid (weak acid) and sodium ethanoate (its conjugate base salt).
- [D] Sodium hydroxide and sodium ethanoate.

45 [EASY] Oxidation of Alkylbenzenes to Benzoic Acid — 9701/11/M/J/220/Q45**[1]**

What is the organic product when methylbenzene (toluene), ethylbenzene, or propylbenzene is refluxed with hot concentrated alkaline KMnO_4 followed by acidification?

- [A] Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$ (regardless of the length of the alkyl side chain, provided a benzylic hydrogen is present).
- [B] Benzaldehyde.
- [C] Phenol.
- [D] Benzophenone.

46 [HARD] Acidity: Propanoic vs Propenoic vs Propynoic Acid — 9701/12/M/J/221/Q46**[1]**

Which sequence ranks these three-carbon acids in order of INCREASING acid strength (lowest K_a to highest K_a)?

- [A] Propynoic acid < Propenoic acid < Propanoic acid
- [B] Propenoic acid < Propanoic acid < Propynoic acid
- [C] All three acids have identical pK_a values.
- [D] Propanoic acid < Propenoic acid < Propynoic acid

47 [EASY] Thermal Decarboxylation of 1,3-Dicarboxylic Acids — 9701/13/M/J/222/Q47**[1]**

What reaction occurs when propanedioic acid (malonic acid, $\text{HOOC}-\text{CH}_2-\text{COOH}$) is heated gently above 140°C ?

- [A] Formation of a cyclic anhydride.
- [B] Thermal decarboxylation via a 6-membered cyclic transition state to yield ethanoic acid (CH_3COOH) and carbon dioxide (CO_2).
- [C] Dehydration to propyne.
- [D] Oxidation to oxalic acid.

48 [HARD] Synthesis of Benzoyl Chloride — 9701/11/M/J/223/Q48**[1]**

What reagent is used to convert benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) cleanly into benzoyl chloride ($\text{C}_6\text{H}_5\text{COCl}$)?

- [A] Concentrated hydrochloric acid.
- [B] Chlorine gas in sunlight.
- [C] Phosphorus(V) chloride, PCl_5 (or thionyl chloride, SOCl_2 , under reflux).
- [D] Sodium chloride.

49 [EASY] Reaction of Carboxylic Acids with Calcium Carbonate — 9701/12/M/J/224/Q49**[1]**

What are the observed products when dilute ethanoic acid reacts with marble chips (calcium carbonate, CaCO_3)?

- [A] Vigorous explosion of hydrogen gas.
- [B] Steady effervescence of CO_2 gas and dissolution of the solid to form a colorless solution of calcium ethanoate, $(\text{CH}_3\text{COO})_2\text{Ca}$.
- [C] A bright yellow precipitate of sulfur.
- [D] No reaction occurs because ethanoic acid is too weak.

50 [HARD] Proximity Effect in Dicarboxylic Acid Dissociation — 9701/13/M/J/220/Q50**[1]**

Why is the first acid dissociation constant (K_{a1}) of ethanedioic acid (oxalic acid, $pK_{a1} = 1.25$) significantly higher than that of ethanoic acid ($pK_a = 4.76$)?

- [A] The second unionized $-COOH$ group exerts a strong electron-withdrawing ($-I$) effect on the adjacent carboxylate group, stabilizing the mono-anion.
- [B] Ethanedioic acid is an inorganic mineral acid.
- [C] Oxalic acid forms stronger London dispersion forces.
- [D] The second proton prevents dissociation.

51 [EASY] Soap Micelle Structure — 9701/11/M/J/221/Q51**[1]**

How do sodium salts of long-chain fatty acids (soaps, e.g. sodium stearate, $C_{17}H_{35}COONa$) remove non-polar grease from fabrics in water?

- [A] Soap chemically reacts with grease to form carbon dioxide.
- [B] Soap dissolves fabric to release dirt.
- [C] Soap freezes the grease into solid crystals.
- [D] The hydrophobic hydrocarbon tails dissolve in the non-polar grease droplet, while the hydrophilic carboxylate heads ($-COO^-$) face outward into water, forming suspended spherical micelles.

52 [HARD] Formation of Ammonium Salts of Carboxylic Acids — 9701/12/M/J/222/Q52**[1]**

What compound is formed when gaseous ammonia is bubbled through glacial ethanoic acid?

- [A] Ethanamide directly
- [B] Ethanenitrile
- [C] Ammonium ethanoate, $CH_3COO^- NH_4^+$ (a white crystalline salt)
- [D] Methylamine

53 [EASY] Thermal Conversion of Ammonium Salts to Amides — 9701/13/M/J/223/Q53**[1]**

How is ethanamide synthesized from solid ammonium ethanoate in the laboratory?

- [A] Heating solid ammonium ethanoate strongly: it undergoes thermal dehydration to yield ethanamide and water vapor ($CH_3COONH_4 \rightarrow CH_3CONH_2 + H_2O$).
- [B] Adding concentrated sulfuric acid at $0^\circ C$.
- [C] Dissolving in aqueous sodium hydroxide.
- [D] Exposing to ultraviolet light.

54 [HARD] Acidic Hydrogens of Acyl Chlorides vs Carbonyls — 9701/11/M/J/224/Q54**[1]**

Why are α -hydrogens in ethanoyl chloride (CH_3COCl) even more acidic than those in ethanal (CH_3CHO)?

- [A] Chlorine donates protons directly.
- [B] Ethanoyl chloride has no α -hydrogens.
- [C] The highly electronegative chlorine atom pulls electron density via the $-I$ effect, further polarizing the $C-H$ bonds and stabilizing the conjugate base.
- [D] Ethanoyl chloride is an ionic base.

55 [EASY] Dicarboxylic Acid Decarboxylation vs Mono-carboxylic — 9701/12/M/J/220/Q55**[1]**

Why does 1,1-ethanedicarboxylic acid (methylmalonic acid) decarboxylate easily upon heating to $150^\circ C$, while propanoic acid is thermally stable above $300^\circ C$?

- [A] Propanoic acid is an ionic solid.
- [B] Gem-dicarboxylic acids decarboxylate via a low-energy 6-membered cyclic hydrogen-transfer mechanism, whereas mono-carboxylic acids require breaking strong $C-C$ and $C-H$ bonds.
- [C] Methylmalonic acid is an explosive.
- [D] Decarboxylation only occurs in alkanes.

SUBTOPIC 18.2 — ESTERS: FORMATION, NAMING & HYDROLYSIS (Q56 – Q100)

56 [EASY] Esterification Reaction Reagents and Catalyst — 9701/11/M/J/23/Q30**[1]**

What reagents, catalyst, and conditions are used to prepare ethyl ethanoate from ethanoic acid and ethanol?

- [A] Mixing with aqueous sodium hydroxide at room temperature.
- [B] Heating with acidified potassium dichromate(VI).
- [C] Bubbling chlorine gas through the mixture in UV light.
- [D] Heating under reflux with a catalytic amount of concentrated sulfuric acid (H_2SO_4).

57 [EASY] Systematic IUPAC Naming of Esters — 9701/12/M/J/23/Q30**[1]**

What is the systematic IUPAC name for the ester with structural formula $\text{CH}_3\text{CH}_2\text{-COO-CH}(\text{CH}_3)_2$?

- [A] Propyl ethanoate
- [B] Ethyl butanoate
- [C] 1-methylethyl propanoate (or isopropyl propanoate)
- [D] 2-methylpropyl ethanoate

58 [HARD] Acid-Catalyzed vs Base-Promoted Hydrolysis of Esters — 9701/13/M/J/23/Q30**[1]**

Why does alkaline hydrolysis (saponification) of an ester go to 100% completion (irreversible), whereas acid-catalyzed hydrolysis reaches an equilibrium mixture?

- [A] In alkaline hydrolysis, the carboxylic acid formed is immediately deprotonated by base into a resonance-stabilized carboxylate anion (RCOO^-), which has no electrophilic center and cannot be attacked by the alcohol.
- [B] The alcohol evaporates completely in alkali.
- [C] Acid catalysts destroy ester molecules.
- [D] Alkaline hydrolysis produces hydrogen gas that escapes.

59 [MEDIUM] Saponification of Natural Triglycerides to Soap — 9701/11/O/N/23/Q30**[1]**

What products are obtained when a vegetable oil (a triester of glycerol and fatty acids) is boiled with aqueous sodium hydroxide?

- [A] Ethanol and sodium carbonate.
- [B] Ethanoic acid and sodium propoxide.
- [C] Biodiesel and methane gas.
- [D] Propane-1,2,3-triol (glycerol) and sodium salts of long-chain fatty acids (soap).

60 [HARD] Hydrolysis Products of Methyl Benzoate — 9701/11/M/J/220/Q60**[1]**

What organic products are formed when methyl benzoate ($\text{C}_6\text{H}_5\text{COOCH}_3$) is refluxed with dilute sodium hydroxide, followed by acidification with dilute hydrochloric acid?

- [A] Phenol and ethanoic acid
- [B] Benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) and methanol (CH_3OH)
- [C] Benzaldehyde and methanoic acid
- [D] Benzene and methyl ethanoate

61 [EASY] Deducing Reactants from Ester Structure — 9701/12/M/J/221/Q61**[1]**

Which carboxylic acid and alcohol were used to synthesize the fruity-smelling ester butyl methanoate, $\text{HCOOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$?

- [A] Butanoic acid and methanol
- [B] Methanoic acid (HCOOH) and butan-1-ol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$)
- [C] Ethanoic acid and propan-1-ol
- [D] Methanal and butan-1-ol

62 [HARD] Ester Isomerism: Structural Isomers of C₃H₆O₂ — 9701/13/M/J/222/Q62**[1]**What are the two isomeric esters that have the molecular formula C₃H₆O₂?

- [A] Propyl methanoate and methyl propanoate
- [B] Ethyl ethanoate and methanoic acid
- [C] Dimethyl carbonate and propanal
- [D] Methyl ethanoate (CH₃COOCH₃) and ethyl methanoate (HCOOCH₂CH₃)

63 [EASY] Ester Isomers of C₄H₈O₂ — 9701/11/M/J/223/Q63**[1]**How many isomeric esters have the molecular formula C₄H₈O₂?

- [A] 4 (ethyl ethanoate, methyl propanoate, propyl methanoate, and isopropyl methanoate)
- [B] 2
- [C] 3
- [D] 5

64 [HARD] Volatility and Intermolecular Forces in Esters — 9701/12/M/J/224/Q64**[1]**

Why do esters have significantly lower boiling points than carboxylic acids of identical molecular mass (e.g. ethyl methanoate bp 54 °C vs propanoic acid bp 141 °C)?

- [A] Esters are completely non-polar.
- [B] Propanoic acid contains covalent network bonds.
- [C] Esters lack an O–H bond and cannot form intermolecular hydrogen bonds with each other, relying only on weaker permanent dipole–dipole attractions.
- [D] Esters decompose below 50 °C.

65 [EASY] Water Solubility of Low-Molecular-Weight Esters — 9701/13/M/J/220/Q65**[1]**Why are small esters like methyl ethanoate moderately soluble in water (~25–33 g per 100 cm³)?

- [A] The lone pairs on the ester carbonyl and alkoxy oxygen atoms can accept hydrogen bonds from water molecules.
- [B] Esters react with water to form ionic salts.
- [C] Methyl ethanoate is completely ionic.
- [D] Water cannot dissolve polar liquids.

66 [HARD] Equilibrium Constant Expression for Esterification — 9701/11/M/J/221/Q66**[1]**What is the equilibrium constant expression, K_c, for the esterification of ethanoic acid with ethanol: CH₃COOH + C₂H₅OH ⇌ CH₃COOC₂H₅ + H₂O?

- [A] $K_c = [\text{CH}_3\text{COOC}_2\text{H}_5] / ([\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}])$ (water is omitted)
- [B] $K_c = [\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}] / [\text{CH}_3\text{COOC}_2\text{H}_5]$
- [C] $K_c = [\text{CH}_3\text{COOC}_2\text{H}_5] / [\text{H}_2\text{O}]$
- [D] $K_c = [\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}] / ([\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}])$ (water MUST be included because it is a reactant/product in a non-aqueous homogeneous mixture)

67 [EASY] Value of K_c for Ethanoic Acid and Ethanol — 9701/12/M/J/222/Q67**[1]**When equimolar amounts of ethanoic acid and ethanol react, approximately 2/3 of each reactant is converted to ester at equilibrium. What is the approximate value of K_c?

- [A] 1.0
- [B] Approximately 4.0
- [C] 0.25
- [D] 10.0

68 [HARD] Yield Improvement: Le Chatelier's Principle in Esterification — 9701/13/M/J/223/Q68

[1]

How can the percentage conversion of carboxylic acid to ester be increased beyond the normal 67% equilibrium yield?

- [A] Cool the reaction mixture to 0 °C.
- [B] Add sodium hydroxide solution.
- [C] Use a large excess of the cheaper alcohol, or continually remove water by adding anhydrous drying agents or using a Dean-Stark water trap.
- [D] Carry out the reaction in an open beaker with water.

69 [EASY] Isotopic Labeling Proof of Esterification Mechanism — 9701/11/M/J/224/Q69

[1]

When ethanoic acid is reacted with methanol labeled with the oxygen-18 isotope ($\text{CH}_3\text{-}^{18}\text{OH}$), which product contains the ^{18}O isotope?

- [A] Water contains the ^{18}O isotope.
- [B] The ester (methyl ethanoate) contains the ^{18}O isotope, proving that the C-OH bond of the carboxylic acid breaks and the C- ^{18}O bond of the alcohol remains intact.
- [C] Both products contain equal ^{18}O .
- [D] The ^{18}O is released as oxygen gas.

70 [HARD] Mechanism of Acid-Catalyzed Esterification — 9701/12/M/J/220/Q70

[1]

In the mechanism of esterification, what is the initial step that makes the carboxylic acid reactive towards nucleophilic attack by the alcohol?

- [A] Protonation of the carbonyl oxygen atom by H^+ , which increases the partial positive charge (δ^+) on the carbonyl carbon.
- [B] Protonation of the alcohol -OH group.
- [C] Deprotonation of the carboxylic acid to a carboxylate anion.
- [D] Homolytic cleavage of the C-O bond.

71 [EASY] Purification of Liquid Ester: Removal of Acid Impurities — 9701/13/M/J/221/Q71

[1]

After esterification, the reaction mixture is poured into aqueous sodium carbonate. What is the visual observation and chemical purpose?

- [A] A dense black precipitate of carbon forms.
- [B] The solution turns bright blue.
- [C] Steam is evolved.
- [D] Vigorous effervescence of CO_2 gas as unreacted ethanoic acid and sulfuric acid are neutralized into soluble sodium salts in the aqueous layer.

72 [HARD] Drying the Isolated Ester Layer — 9701/11/M/J/222/Q72

[1]

Which drying agent is added to the separated organic ester layer to remove traces of dissolved water before final distillation?

- [A] Concentrated sulfuric acid
- [B] Solid sodium hydroxide pellets
- [C] Anhydrous magnesium sulfate (MgSO_4) or anhydrous calcium chloride (CaCl_2)
- [D] Phosphorus(V) oxide

73 [EASY] Fractional Distillation of Synthesized Ester — 9701/12/M/J/223/Q73

[1]

Why must the dry ester be purified by distillation, and what temperature fraction is collected?

- [A] To separate the ester from unreacted alcohol; collect the fraction boiling within $\pm 2^\circ\text{C}$ of the ester's literature boiling point (77°C for ethyl ethanoate).
- [B] To burn off excess oxygen.
- [C] To convert the ester into an acid.
- [D] To freeze the ester into crystals.

74 [HARD] Flavorings and Fragrances of Common Esters — 9701/13/M/J/224/Q74

[1]

Which ester is responsible for the characteristic artificial flavor and aroma of bananas and pears?

- [A] Methyl methanoate
- [B] Ethyl methanoate
- [C] 3-methylbutyl ethanoate (isoamyl acetate)
- [D] Octyl ethanoate

75 [EASY] Aroma of Ethyl Butanoate — 9701/11/M/J/220/Q75

[1]

What commercial fruit aroma is associated with the ester ethyl butanoate ($\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$)?

- [A] Banana
- [B] Pineapple
- [C] Apple
- [D] Almond

76 [HARD] Aroma of Octyl Ethanoate — 9701/12/M/J/221/Q76

[1]

What fruit aroma is characteristic of octyl ethanoate ($\text{CH}_3\text{COOCH}_2(\text{CH}_2)_6\text{CH}_3$)?

- [A] Raspberry
- [B] Grape
- [C] Lemon
- [D] Orange

77 [EASY] Lactones: Cyclic Esters — 9701/13/M/J/222/Q77

[1]

What is a lactone, and from what type of starting material is it formed by intramolecular esterification?

- [A] A cyclic ether formed from diols.
- [B] An ester containing nitrogen atoms.
- [C] A cyclic ester, formed by the intramolecular condensation of a hydroxy-carboxylic acid (e.g. 4-hydroxybutanoic acid forming a 5-membered lactone).
- [D] A polymer of ethene.

78 [HARD] Hydrolysis of Polyesters (PET) — 9701/11/M/J/223/Q78

[1]

What are the monomer products obtained upon complete alkaline hydrolysis of poly(ethylene terephthalate) (PET plastic)?

- [A] Benzene-1,4-dicarboxylate salt (terephthalate) and ethane-1,2-diol (ethylene glycol)
- [B] Ethanol and benzoic acid
- [C] Ethene and phthalic acid
- [D] Glucose and water

79 [EASY] Saponification Number Definition — 9701/12/M/J/224/Q79

[1]

What is the saponification value of a fat or oil, and what information does it provide about molecular size?

- [A] The volume of oxygen consumed during combustion.
- [B] The percentage of water in the fat.
- [C] The melting point of the soap.
- [D] The number of milligrams of KOH required to completely saponify 1.0 g of the fat; higher saponification values indicate shorter average fatty acid chain lengths (lower Mr).

80 [HARD] Biodiesel Synthesis via Transesterification — 9701/13/M/J/220/Q80

[1]

In the production of biodiesel fuel from soybean or canola oil, what alcohol and catalyst are used to perform transesterification?

- [A] Ethanol with concentrated hydrochloric acid.
- [B] Methanol with a strong base catalyst (potassium hydroxide or sodium methoxide).
- [C] Propan-1-ol with nickel catalyst.
- [D] Phenol with sulfuric acid.

81 [EASY] Why Biodiesel is Superior to Raw Vegetable Oil in Engines — 9701/11/M/J/221/Q81

[1]

Why must vegetable oil be converted into methyl esters (biodiesel) before being used in standard modern diesel engines?

- [A] Vegetable oil does not burn.
- [B] Vegetable oil has an extremely high viscosity that clogs fuel injectors and causes carbon build-up; transesterification breaks the heavy triglyceride into low-viscosity, volatile methyl esters.
- [C] Vegetable oil is an explosive solid.
- [D] Raw oil produces chlorine gas.

82 [HARD] Hydrolysis of Phenyl Esters — 9701/12/M/J/222/Q82

[1]

What are the products when phenyl ethanoate ($\text{CH}_3\text{COOC}_6\text{H}_5$) is heated under reflux with excess aqueous sodium hydroxide?

- [A] Ethanoic acid and phenol only
- [B] Benzene and sodium ethanoate
- [C] Benzoic acid and ethanol
- [D] Sodium ethanoate (CH_3COONa) and sodium phenoxide ($\text{C}_6\text{H}_5\text{ONa}$)

83 [EASY] Hydrolysis of Esters with ^{18}O Labeled Water — 9701/13/M/J/223/Q83

[1]

When ethyl ethanoate is hydrolyzed with water enriched with oxygen-18 ($\text{H}_2\text{-}^{18}\text{O}$) in acid, which product contains the ^{18}O label?

- [A] Ethanoic acid contains the ^{18}O atom ($\text{CH}_3\text{CO-}^{18}\text{OH}$), while ethanol contains ordinary ^{16}O .
- [B] Ethanol contains ^{18}O .
- [C] Both contain ^{18}O equally.
- [D] Carbon dioxide contains ^{18}O .

84 [HARD] Aspirin Synthesis via Esterification — 9701/11/M/J/224/Q84

[1]

Aspirin (acetylsalicylic acid) is synthesized by esterifying 2-hydroxybenzoic acid (salicylic acid). Which functional group is esterified, and what reagent is preferred?

- [A] The carboxylic acid group is esterified using methanol.
- [B] The benzene ring is brominated.
- [C] The phenolic $-\text{OH}$ group is esterified using ethanoic anhydride, $(\text{CH}_3\text{CO})_2\text{O}$.
- [D] The molecule is reduced with NaBH_4 .

85 [EASY] Hydrolysis of Aspirin in the Body — 9701/12/M/J/220/Q85

[1]

When aspirin (acetylsalicylic acid) is absorbed in the alkaline environment of the small intestine, what hydrolysis products are formed?

- [A] Salicylate ions (2-hydroxybenzoate) and ethanoate ions (acetate)
- [B] Phenol and carbonic acid
- [C] Benzoic acid and methanol
- [D] Ethanal and benzene

86 [HARD] Paracetamol Structure vs Aspirin — 9701/13/M/J/221/Q86**[1]**

How does paracetamol differ from an ester in its functional grouping?

- [A]** Paracetamol is a carboxylic acid.
- [B]** Paracetamol is an alkene.
- [C]** Paracetamol contains no aromatic rings.
- [D]** Paracetamol contains an amide group ($-\text{NH}-\text{COCH}_3$) attached to a phenol ring, whereas aspirin contains an ester group ($-\text{O}-\text{COCH}_3$).

87 [EASY] Reactivity of Esters Towards Nucleophiles — 9701/11/M/J/222/Q87**[1]**

Why are esters significantly less reactive towards nucleophilic attack than aldehydes and ketones?

- [A]** Esters are non-polar.
- [B]** The non-bonding lone pair on the alkoxy oxygen ($-\text{OR}'$) delocalises into the carbonyl pi orbital via resonance, reducing the δ^+ charge on the carbonyl carbon.
- [C]** Esters contain no double bonds.
- [D]** The carbonyl group in esters is linear.

88 [HARD] Reduction of Esters with LiAlH_4 — 9701/12/M/J/223/Q88**[1]**What two alcohol products are formed when ethyl propanoate is reduced by LiAlH_4 in dry ether followed by acidification?

- [A]** Propanoic acid and ethane.
- [B]** Pentan-1-ol only.
- [C]** Propan-1-ol (from the acyl group) and ethanol (from the alkoxy group).
- [D]** Propan-2-ol and ethanol.

89 [EASY] Esters as Solvents in Industry — 9701/13/M/J/224/Q89**[1]**

Why is ethyl ethanoate widely used as a solvent in nail polish removers, glues, and model airplane cements?

- [A]** It is an inflammable mineral oil.
- [B]** It has moderate polarity, dissolves diverse organic solutes (nitrocellulose, polymers), has low toxicity, and evaporates rapidly without greasy residue.
- [C]** It reacts chemically with nails to strengthen them.
- [D]** It turns into solid plastic upon exposure to air.

90 [HARD] Wax Chemistry: Long-Chain Esters — 9701/11/M/J/220/Q90**[1]**

What is the chemical composition of natural beeswax (myricyl palmitate)?

- [A]** A high-molecular-weight ester formed from a long-chain carboxylic acid (palmitic acid, C_{16}) and a long-chain alcohol (myricyl alcohol, C_{30}).
- [B]** A polymer of ethene.
- [C]** A mixture of liquid alkanes.
- [D]** A salt of sodium and calcium.

91 [EASY] Condensation Polymerisation: Terylene / Dacron — 9701/12/M/J/221/Q91**[1]**

What are the two co-monomers used in the industrial condensation polymerisation to manufacture the polyester Terylene (Dacron)?

- [A]** Ethene and propene
- [B]** 1,6-diaminohexane and hexanedioic acid
- [C]** Phenol and methanal
- [D]** Benzene-1,4-dicarboxylic acid (terephthalic acid) and ethane-1,2-diol (ethylene glycol)

92 [HARD] Deducing Ester Structure from Mass Spectrum Base Peak — 9701/13/M/J/222/Q92**[1]**

In the mass spectrum of methyl ethanoate ($\text{CH}_3\text{COOCH}_3$, $M_r = 74.0$), an intense base peak is observed at $m/z = 43$. What fragment ion is responsible for this peak?

- [A] The methoxy cation, $[\text{CH}_3\text{O}]^+$
- [B] The ethyl cation, $[\text{C}_2\text{H}_5]^+$
- [C] The ethanoyl / acylium cation, $[\text{CH}_3\text{-C=O}]^+$ ($m/z = 15 + 28 = 43$)
- [D] The carboxylate radical, $[\text{COO}]^+$

93 [EASY] Deducing Ester Structure from NMR Singlet — 9701/11/M/J/223/Q93**[1]**

An unknown ester of formula $\text{C}_3\text{H}_6\text{O}_2$ displays two sharp singlets in its ^1H NMR spectrum with peak area ratio 3 : 3. What is the ester?

- [A] Methyl ethanoate, $\text{CH}_3\text{COOCH}_3$ (one methyl bonded to carbonyl, one methyl bonded to oxygen; neither has adjacent hydrogens).
- [B] Ethyl methanoate.
- [C] Propanoic acid.
- [D] Methoxyethene.

94 [HARD] Ester Hydrolysis Kinetics — 9701/12/M/J/224/Q94**[1]**

Why is the base-promoted hydrolysis of an ester pseudo-first-order when carried out in a large excess of aqueous sodium hydroxide?

- [A] Because esters only react with one molecule of base.
- [B] Because the reaction is unimolecular ($\text{S}_{\text{N}}1$).
- [C] Because $[\text{OH}^-]$ remains virtually constant throughout the reaction, the rate depends only on [ester]: $\text{Rate} = k[\text{ester}]$.
- [D] Because water is consumed.

95 [EASY] Fats vs Oils: Saturated vs Unsaturated Esters — 9701/13/M/J/220/Q95**[1]**

Why are animal fats (such as lard and tallow) solids at room temperature while plant oils (such as olive and sunflower oil) are liquids?

- [A] Fats contain ionic bonds.
- [B] Fats contain mostly saturated fatty acid chains that pack closely together, maximizing London dispersion forces; plant oils contain cis-unsaturated double bonds that create kinks, preventing close molecular packing.
- [C] Plant oils contain water.
- [D] Fats have lower molecular masses.

96 [HARD] Hydrogenation of Vegetable Oils: Trans Fats — 9701/11/M/J/221/Q96**[1]**

What undesirable byproduct can be formed during the partial catalytic hydrogenation of vegetable oils into margarine?

- [A] Toxic cyanides.
- [B] Glycerol trichloride.
- [C] Carboxylic acid polymers.
- [D] Trans-fatty acid isomers, formed by reversible double-bond isomerisation on the nickel catalyst surface.

97 [EASY] Ester Cleavage by Ammonia (Ammonolysis) — 9701/12/M/J/222/Q97**[1]**

What products are formed when ethyl ethanoate is shaken with concentrated aqueous ammonia at room temperature?

- [A] Ethylamine and ethanoic acid
- [B] Aminoethane and water
- [C] Ethanamide (CH_3CONH_2) and ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)
- [D] Ethanenitrile and ethanol

98 [HARD] Reaction of Esters with Grignard Reagents — 9701/13/M/J/223/Q98**[1]**

What class of alcohol is synthesized when an ester (other than methanoate) reacts with two equivalents of a Grignard reagent followed by acid hydrolysis?

- [A] A tertiary alcohol with two identical alkyl groups, $R-C(OH)(R')_2$
- [B] A primary alcohol
- [C] A secondary alcohol
- [D] A ketone

99 [EASY] Ester Distinction from Carboxylic Acids by Spectroscopy — 9701/11/M/J/224/Q99**[1]**

How does the infrared spectrum of an ester differ definitively from that of its carboxylic acid isomer?

- [A] The ester has no carbonyl absorption.
- [B] The carboxylic acid has no C–O absorption.
- [C] Both spectra are completely indistinguishable.
- [D] The ester displays a sharp carbonyl peak at $\sim 1740\text{ cm}^{-1}$ and C–O peaks at $1100\text{--}1250\text{ cm}^{-1}$, but completely lacks the very broad, ragged O–H band ($2500\text{--}3300\text{ cm}^{-1}$) present in carboxylic acids.

100 [HARD] Saponification of Methyl Salicylate — 9701/12/M/J/220/Q100**[1]**

What products are obtained when methyl salicylate (oil of wintergreen, 2-HO-C₆H₄-COOCH₃) is heated under reflux with excess aqueous sodium hydroxide?

- [A] Salicylic acid and sodium methoxide.
- [B] Disodium salicylate (2-NaO-C₆H₄-COONa) and methanol (CH₃OH).
- [C] Phenol, sodium carbonate, and methanol.
- [D] Methyl benzoate and water.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101 [HARD] [HIGH-FREQUENCY EXAM REPEAT]** HF 1: Inductive Effect Ranking on Carboxylic Acid Strength — 9701/11/M/J/22/Q29**[1]**

Consider the following four carboxylic acids:

1: CH₃COOH (ethanoic acid); 2: CH₂ClCOOH (chloroethanoic acid); 3: CHCl₂COOH (dichloroethanoic acid); 4: CH₂FCOOH (fluoroethanoic acid)

Which sequence correctly lists these acids in order of INCREASING acid strength (lowest K_a to highest K_a)?

- [A] $1 < 4 < 2 < 3$
- [B] $3 < 4 < 2 < 1$
- [C] $1 < 2 < 4 < 3$
- [D] $4 < 3 < 2 < 1$

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 2: Quantitative Saponification Calculation — 9701/12/M/J/21/Q29**[1]**

A sample of 0.050 mol of an unknown ester of formula C₅H₁₀O₂ is heated with 100 cm³ of 1.00 mol dm⁻³ aqueous sodium hydroxide. After complete hydrolysis, the excess sodium hydroxide requires 50.0 cm³ of 1.00 mol dm⁻³ hydrochloric acid for neutralization.

How many ester functional groups were present in one molecule of the ester, and was it a mono-ester or di-ester?

- [A] 1 ester group (it is a mono-ester); consumed exactly 0.050 mol of NaOH.
- [B] 2 ester groups (it is a di-ester).
- [C] 3 ester groups (it is a tri-ester).
- [D] 0 ester groups.

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 3: Identification of Ester from Hydrolysis Products — 9701/13/O/N/22/Q29**[1]**

An ester X has the molecular formula $C_5H_{10}O_2$. When X is boiled with aqueous sodium hydroxide followed by acidification, two organic compounds, P and Q, are formed.

- Compound P turns acidified potassium dichromate(VI) from orange to green upon warming and gives a yellow precipitate with alkaline aqueous iodine.
- Compound Q produces vigorous effervescence of CO_2 with sodium hydrogencarbonate.

What is the structural formula of ester X?

- [A] Propyl ethanoate, $CH_3-COO-CH_2CH_2CH_3$
[B] Ethyl propanoate, $CH_3CH_2-COO-CH_2CH_3$
[C] Methyl butanoate, $CH_3CH_2CH_2-COO-CH_3$
[D] 1-methylethyl ethanoate (isopropyl ethanoate), $CH_3-COO-CH(CH_3)_2$

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 4: Distinction Between Carboxylic Acids and Esters — 9701/11/F/M/22/Q29**[1]**

Two structural isomers, W and Z, both have the molecular formula $C_3H_6O_2$.

- Isomer W reacts with sodium carbonate to evolve a gas that turns limewater cloudy.
- Isomer Z does NOT react with sodium carbonate, but when boiled with aqueous NaOH and acidified, produces a single alcohol that gives a positive iodoform test.

What are the structural formulas of isomers W and Z?

- [A] W is ethyl methanoate; Z is propanoic acid.
[B] W is propanoic acid (CH_3CH_2COOH); Z is methyl ethanoate (CH_3COOCH_3).
[C] W is propanoic acid; Z is ethyl methanoate.
[D] W is methyl ethanoate; Z is propanoic acid.

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 5: Stoichiometry of Ester Combustion — 9701/12/O/N/21/Q30**[1]**

What volume of carbon dioxide gas (measured at RTP) is produced by the complete combustion of 0.10 mol of ethyl propanoate, $C_5H_{10}O_2$?

- [A] 12.0 dm^3 of CO_2 (0.50 mol)
[B] 24.0 dm^3 of CO_2
[C] 4.8 dm^3 of CO_2
[D] 2.4 dm^3 of CO_2

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 6: Mechanism of Acyl Chloride Conversion to Esters — 9701/13/M/J/22/Q29**[1]**

Ethanoyl chloride reacts vigorously with ethanol at room temperature to form ethyl ethanoate and HCl gas.

What is the reaction mechanism for this conversion?

- [A] Electrophilic substitution into the acyl chain.
[B] Free radical substitution.
[C] Nucleophilic addition-elimination (addition of ethanol to the carbonyl carbon followed by elimination of chloride ion and a proton).
[D] Electrophilic addition across the $C=O$ double bond.

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 7: Relative Acid Strength Comparison Across Different Chemical Families — 9701/11/O/N/21/Q29**[1]**

Which sequence correctly ranks the following organic compounds in order of DECREASING acid strength (most acidic to least acidic)?

- [A] Ethanoic acid > Trichloroethanoic acid > Ethanol > Phenol
[B] Trichloroethanoic acid > Ethanoic acid > Phenol > Ethanol
[C] Phenol > Trichloroethanoic acid > Ethanoic acid > Ethanol
[D] Ethanol > Phenol > Ethanoic acid > Trichloroethanoic acid

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 8: Complete Oxidation of Unknown Diol to Dicarboxylic Acid — 9701/12/F/M/23/Q30**[1]**

0.10 mol of a dihydric alcohol (diol) is completely oxidized by heating under reflux with acidified potassium dichromate(VI). The resulting dicarboxylic acid requires exactly 40.0 cm³ of 5.00 mol dm⁻³ NaOH for complete neutralization.

What structural feature must the starting diol possess?

- [A] One hydroxyl group is primary and one is secondary.
- [B] Both hydroxyl groups are tertiary.
- [C] The diol must be a cyclic compound.
- [D] Both hydroxyl groups must be primary (–CH₂OH) alcohols.

109 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 9: Predicting Products of Cross-Esterification with Methanoic Acid — 9701/13/M/J/23/Q30**[1]**

Which ester will produce a silver mirror when warmed with Tollens' reagent?

- [A] Methyl methanoate, HCOOCH₃
- [B] Ethyl ethanoate, CH₃COOCH₂CH₃
- [C] Methyl propanoate, CH₃CH₂COOCH₃
- [D] Phenyl ethanoate, CH₃COOC₆H₅

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 10: Deductive Multi-Step Synthetic Yield Calculation — 9701/11/M/J/23/Q31**[1]**

Consider the two-step synthesis of ethyl ethanoate from ethene:

Step 1: Ethene + steam → Ethanol (Yield = 80.0%)

Step 2: Ethanol + ethanoic acid → Ethyl ethanoate (Yield = 75.0%)

What is the overall percentage yield of ethyl ethanoate from ethene, and how many moles of ethene are required to produce 0.60 mol of ethyl ethanoate?

- [A] Overall yield = 77.5%; requires 0.77 mol of ethene.
- [B] Overall yield = 50.0%; requires 1.20 mol of ethene.
- [C] Overall yield = 60.0%; requires 1.00 mol of ethene.
- [D] Overall yield = 60.0%; requires 0.60 mol of ethene.

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------

COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. Carboxylic acids possess both a hydrogen bond donor (–OH) and a hydrogen bond acceptor (C=O). Two molecules pair up through two reciprocal hydrogen bonds, forming a stable 8-membered cyclic dimer. This effective doubling of molecular mass and strong intermolecular attraction raises the boiling point significantly above that of isomeric or equal-Mr alcohols.	[1]
2	D	Option D is correct. In the alkoxide ion (RO [–]) formed from an alcohol, the full negative charge is localized on a single oxygen atom and destabilized by the +I effect of the alkyl group. In contrast, the carboxylate anion (RCOO [–]) undergoes resonance delocalisation: the unshared electron pair on the deprotonated oxygen overlaps with the carbonyl pi system, distributing the negative charge equally across both oxygen atoms (–1/2 on each). Both C–O bonds become identical in length (0.127 nm), imparting substantial thermodynamic stability to the anion.	[1]
3	A	Option A is correct. Chlorine atoms are highly electronegative and exert a powerful electron-withdrawing inductive effect (–I) through sigma bonds. As more chlorine atoms are introduced onto the alpha-carbon, electron density is pulled away from the carboxylate group, dispersing the negative charge across the ion. Dispersal of negative charge stabilizes the conjugate base anion, shifting the dissociation equilibrium to the right and increasing K _a (pK _a : CH ₃ COOH 4.76 > CH ₂ ClCOOH 2.86 > CHCl ₂ COOH 1.29 > CCl ₃ COOH 0.65).	[1]
4	C	Option C is correct. Carboxylic acids are strong enough acids (pK _a ~ 4.8) to protonate carbonate and hydrogencarbonate ions, evolving carbon dioxide gas with vigorous effervescence: RCOOH + NaHCO ₃ → RCOONa + CO ₂ (g) + H ₂ O. Alcohols (pK _a ~ 16) and phenols (pK _a ~ 10) are far too weakly acidic to react with NaHCO ₃ /Na ₂ CO ₃ , making effervescence with hydrogencarbonate the definitive diagnostic test for carboxylic acids.	[1]
5	A	Option A is correct. Inductive effects diminish rapidly with distance along a carbon chain, becoming negligible beyond three sigma bonds. In 2-chlorobutanoic acid (CH ₃ CH ₂ CH(Cl)COOH), the chlorine is on the alpha-carbon, exerting the strongest electron withdrawal (–I) on the carboxylate group and giving the highest K _a (pK _a 2.86 vs 3.98 for 3-chloro and 4.52 for 4-chloro).	[1]
6	D	Option D is correct. Alkyl groups release electron density (+I effect). The methyl group pushes electrons toward the carboxylate anion, concentrating the negative charge on oxygen and making it less stable (higher affinity for H ⁺). Methanoic acid has only a hydrogen atom (no +I release), so its conjugate base is more stable.	[1]
7	B	Option B is correct. Acid strength depends on the electronegativity of the halogen atom: F (4.0) > Cl (3.0) > Br (2.8) > I (2.5). Fluorine exerts the strongest –I inductive effect, providing the greatest charge dispersal and stability to the carboxylate ion, making fluoroethanoic acid the strongest acid (pK _a : FCH ₂ COOH 2.59 < ClCH ₂ COOH 2.86 < BrCH ₂ COOH 2.90 < ICH ₂ COOH 3.18).	[1]
8	C	Option C is correct. Carboxylic acids react with metals above hydrogen in the reactivity series (Mg, Zn, Fe): 2RCOOH + M → (RCOO) ₂ M + H ₂ (g). Effervescence is steady (slower than with HCl because carboxylic acids are weakly dissociated).	[1]
9	B	Option B is correct. An acid–base neutralization: 2CH ₃ COOH + CuO(s) → (CH ₃ COO) ₂ Cu(aq) + H ₂ O(l). The black CuO powder dissolves to form a characteristic blue–green solution of copper(II) ethanoate.	[1]
10	A	Option A is correct. The standard neutralization enthalpy of strong acid with strong base is –57.1 kJ/mol (H ⁺ + OH [–] → H ₂ O). Ethanoic acid is weak and largely unionized (~1% dissociated). A small amount of energy (~1.9 kJ/mol) must be absorbed to break the O–H bonds and fully dissociate the acid, making the net enthalpy slightly less exothermic (–55.2 kJ/mol).	[1]
11	D	Option D is correct. LiAlH ₄ is a powerful reducing agent that reduces carboxylic acids directly to primary alcohols in excellent yield: RCOOH + 4[H] → RCH ₂ OH + H ₂ O. (Aldehydes cannot be isolated as they are reduced faster than the parent acid).	[1]
12	C	Option C is correct. Carboxylic acids act as Bronsted acids, protonating the hydride donor to evolve H ₂ gas and forming the carboxylate anion (RCOO [–]). The resonance-stabilized negative charge on the carboxylate group strongly repels incoming nucleophilic hydride ions (:H [–]), rendering reduction impossible with NaBH ₄ .	[1]
13	A	Option A is correct. Reaction: CH ₃ COOH + PCl ₅ → CH ₃ COCl + POCl ₃ + HCl(g). Steamy acidic fumes of HCl are evolved vigorously, and fractional distillation separates ethanoyl chloride (bp 51 °C) from phosphoryl chloride (POCl ₃ , bp 106 °C).	[1]
14	C	Option C is correct. Equation: RCOOH + SOCl ₂ → RCOCl + SO ₂ (g) + HCl(g). Because sulfur dioxide and hydrogen chloride bubble out as gases, the liquid acyl chloride is obtained in high yield and purity without contaminating solid or liquid phosphorus byproducts.	[1]
15	B	Option B is correct. Reaction: 3CH ₃ COOH + PCl ₃ → 3CH ₃ COCl + H ₃ PO ₃ . This reaction requires heating and yields liquid phosphorous acid as a byproduct.	[1]
16	D	Option D is correct. Lower carboxylic acids form extensive hydrogen bonds with water via both the C=O and O–H groups. For longer-chain acids (C >= 6), the non-polar hydrophobic alkyl chain dominates, creating an unfavorable entropy/enthalpy balance for dissolution.	[1]
17	C	Option C is correct. Deprotonation creates two completely equivalent resonance contributors [R–C(=O)–O [–] ↔ R–C(O [–])=O]. The pi electron pair is evenly shared over the O–C–O framework, giving both bonds identical partial double-bond character (bond order 1.5) and an intermediate length of 0.127 nm.	[1]
18	A	Option A is correct. Methanoic acid has structure H–C(=O)OH. The hydrogen attached directly to the carbonyl carbon makes it an aldehyde analog (formyl group). Mild oxidizing agents (Tollens', Fehling's, acidified KMnO ₄) oxidize HCOOH to carbonic acid [H ₂ CO ₃], which dissociates to CO ₂ + H ₂ O: HCOOH + [O] → CO ₂ + H ₂ O. Ethanoic acid has a stable methyl group (CH ₃ –COOH) and cannot be oxidized.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
19	D	Option D is correct. Ethanedioic acid (oxalic acid) is readily oxidized by acidified KMnO_4 : $5(\text{COOH})_2 + 2\text{MnO}_4^- + 6\text{H}^+ \rightarrow 10\text{CO}_2 + 2\text{Mn}^{2+} + 8\text{H}_2\text{O}$. The purple permanganate ion is reduced to colorless Mn^{2+} with vigorous bubbling of CO_2 (used in standard redox titrations at 60 °C).	[1]
20	B	Option B is correct. 1-bromopropane has 3 carbons; butanoic acid has 4 carbons. Lengthening the carbon chain requires nucleophilic substitution with ethanolic KCN under reflux to yield butanenitrile ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$), followed by acid hydrolysis with dilute HCl under reflux: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{COOH} + \text{NH}_4\text{Cl}$.	[1]
21	B	Option B is correct. Grignard carbonation: $\text{C}_6\text{H}_5\text{Br} + \text{Mg} \rightarrow \text{C}_6\text{H}_5\text{MgBr}$. The nucleophilic carbanion C_6H_5^- attacks CO_2 : $\text{C}_6\text{H}_5\text{MgBr} + \text{CO}_2 \rightarrow \text{C}_6\text{H}_5\text{COOMgBr}$. Acid hydrolysis yields benzoic acid: $\text{C}_6\text{H}_5\text{COOMgBr} + \text{H}^+ \rightarrow \text{C}_6\text{H}_5\text{COOH} + \text{Mg}^{2+} + \text{Br}^-$.	[1]
22	D	Option D is correct. The aromatic ring carbon bonded to $-\text{COOH}$ is sp^2 hybridised (33% s-character), which is more electronegative than the sp^3 carbon (25% s-character) of a methyl group. It exerts a weak -I inductive effect, stabilizing the benzoate anion and making benzoic acid slightly stronger than ethanoic acid.	[1]
23	A	Option A is correct. The $-\text{NO}_2$ group is intensely electron-withdrawing through both sigma bonds (-I effect) and the conjugated pi system (-M mesomeric effect). It pulls electron density out of the ring and carboxylate group, stabilizing the 4-nitrobenzoate anion and increasing K_a .	[1]
24	C	Option C is correct. The oxygen atom in $-\text{OCH}_3$ donates an electron lone pair into the aromatic ring via resonance (+M effect). This electron donation pushes electron density toward the carboxylate group, concentrating negative charge and destabilizing the anion, lowering acid strength.	[1]
25	A	Option A is correct. Heating a carboxylate salt with soda lime (a solid mixture of NaOH and CaO) eliminates the carboxyl group as sodium carbonate, yielding an alkane with one fewer carbon atom: $\text{RCOONa} + \text{NaOH} \rightarrow \text{RH} + \text{Na}_2\text{CO}_3$.	[1]
26	D	Option D is correct. Dicarboxylic acids with 4 or 5 carbon atoms (like butanedioic and pentanedioic acid) undergo intramolecular dehydration upon heating, eliminating a molecule of water to form stable, strain-free 5- or 6-membered cyclic anhydrides.	[1]
27	B	Option B is correct. Ethanol oxidizes to ethanal first. Ethanal is very volatile (bp 21 °C) and would escape from an open vessel. Heating under reflux continually condenses ethanal back into the boiling mixture with excess oxidizing agent, forcing complete conversion into ethanoic acid.	[1]
28	C	Option C is correct. Moles of NaOH = $0.0250 \text{ dm}^3 \times 0.100 \text{ mol/dm}^3 = 0.00250 \text{ mol}$. Since the acid is monoprotic: moles of acid = 0.00250 mol. Molar mass $M_r = \text{mass} / \text{moles} = 0.150 \text{ g} / 0.00250 \text{ mol} = 60.0 \text{ g/mol}$. Formula mass of $\text{CH}_3\text{COOH} = 12 + 3 + 12 + 32 + 1 = 60.0$. The acid is ethanoic acid.	[1]
29	B	Option B is correct. Acyl chlorides react vigorously with cold water in an addition-elimination (hydrolysis) reaction: $\text{CH}_3\text{COCl} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{HCl(g)}$. Dense, choking steamy fumes of HCl gas are evolved immediately.	[1]
30	A	Option A is correct. Acyl chlorides react rapidly and irreversibly with alcohols without requiring an acid catalyst or heat: $\text{CH}_3\text{COCl} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3 + \text{HCl(g)}$, yielding the ester propyl ethanoate in high purity.	[1]
31	D	Option D is correct. Reaction: $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$. The nucleophile NH_3 attacks the acyl carbon, expelling Cl^- . A second NH_3 molecule neutralizes the liberated HCl to form solid ammonium chloride smoke.	[1]
32	C	Option C is correct. Nucleophilic addition-elimination of methylamine yields an N-substituted (secondary) amide: $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_3 + \text{CH}_3\text{NH}_3^+ \text{Cl}^-$.	[1]
33	A	Option A is correct. Acyl chlorides (RCOCl) have the most electronegative leaving group (Cl^-) with weak resonance donation, making the carbonyl carbon intensely δ^+ . Anhydrides have two withdrawing carbonyls. Esters have alkoxy resonance donation (+M). Amides have the strongest resonance donation from nitrogen, making the carbon least electrophilic.	[1]
34	C	Option C is correct. Diazomethane is a resonance-stabilized yellow gas in ether that protonates immediately with carboxylic acids, transferring a methyl group and expelling inert N_2 gas to form methyl esters with 100% yield.	[1]
35	B	Option B is correct. Methanoic acid (formic acid, from Latin 'formica' for ant) is found in ant venom and stinging nettle trichomes; ethanoic acid (acetic acid) is produced by bacterial oxidation of wine into vinegar; citric acid is abundant in lemons and oranges.	[1]
36	D	Option D is correct. The enzyme pyruvate dehydrogenase oxidatively decarboxylates pyruvate, eliminating the carboxylate group as carbon dioxide (CO_2) and transferring the remaining two-carbon acetyl fragment to Coenzyme A.	[1]
37	C	Option C is correct. Methanoic (mp 8 °C, bp 101 °C) and ethanoic (mp 17 °C, bp 118 °C) acids are liquids at 20 °C. As chain length increases, London dispersion forces strengthen; stearic acid ($\text{C}_{18}\text{H}_{36}\text{O}_2$, mp 69 °C) is a solid fatty acid.	[1]
38	A	Option A is correct. Pure anhydrous ethanoic acid has a melting point of 16.6 °C (62 °F). On cool days, it solidifies into translucent, ice-like crystals resembling glacier ice, giving rise to the historical name 'glacial' ethanoic acid.	[1]
39	D	Option D is correct. Benzene is non-polar and cannot solvate or separate ions; ethanoic acid remains entirely molecular as neutral hydrogen-bonded dimers. Water has a high dielectric constant and solvates ions through hydrogen bonding, promoting partial ionization into mobile conducting ions.	[1]
40	B	Option B is correct. Nucleophilic substitution by the carboxylate anion on the acyl chloride: $\text{CH}_3\text{COCl} + \text{CH}_3\text{COONa} \rightarrow (\text{CH}_3\text{CO})_2\text{O} + \text{NaCl}$, forming the symmetrical acid anhydride ethanoic anhydride.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
41	B	Option B is correct. Phenol is a weak nucleophile because oxygen's lone pair is delocalised into the benzene ring. Carboxylic acids are insufficiently electrophilic to react with phenol. Acyl chlorides have an intensely electron-deficient carbonyl carbon that reacts smoothly with phenoxide ions ($\text{C}_6\text{H}_5\text{O}^-$) to form phenyl esters (e.g. phenyl ethanoate).	[1]
42	D	Option D is correct. Carboxylic acids exhibit a uniquely broad, jagged O-H stretching band spanning $2500\text{--}3300\text{ cm}^{-1}$ (often overlapping and engulfing the C-H stretches) due to extensive hydrogen-bonded dimerisation, accompanied by a strong, sharp carbonyl ($\text{C}=\text{O}$) absorption at 1710 cm^{-1} .	[1]
43	A	Option A is correct. At equivalence, all ethanoic acid is converted into sodium ethanoate. Ethanoate is the conjugate base of a weak acid and hydrolyzes water: $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$, producing excess hydroxide ions and shifting the equivalence point pH to ~ 8.7 (requiring phenolphthalein indicator).	[1]
44	C	Option C is correct. An acidic buffer contains a weak acid (CH_3COOH) to neutralize added base ($\text{CH}_3\text{COOH} + \text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$) and its conjugate base salt (CH_3COONa) to neutralize added acid ($\text{CH}_3\text{COO}^- + \text{H}^+ \rightarrow \text{CH}_3\text{COOH}$), maintaining a virtually constant pH around $\text{pK}_a \sim 4.76$.	[1]
45	A	Option A is correct. Alkyl side chains attached to a benzene ring are oxidized down to a single benzoic acid group ($-\text{COOH}$) by hot concentrated KMnO_4 , provided the benzylic carbon has at least one C-H bond. All remaining side-chain carbons are cleaved and oxidized to $\text{CO}_2 + \text{H}_2\text{O}$.	[1]
46	D	Option D is correct. Electronegativity of carbon increases with s-character: sp_3 (25% s in propanoic) < sp^2 (33% s in propenoic) < sp (50% s in propynoic). The sp hybridised carbon of propynoic acid ($\text{HC}\equiv\text{C}-\text{COOH}$) exerts the strongest electron-withdrawing ($-I$) effect, stabilizing the carboxylate ion and making it the strongest acid ($\text{pK}_a: 4.87 < 4.25 < 1.84$).	[1]
47	B	Option B is correct. 1,3-dicarboxylic acids (having two $-\text{COOH}$ groups separated by one carbon) readily decarboxylate upon heating via a concerted cyclic 6-membered hydrogen-transfer transition state, eliminating CO_2 gas to produce a mono-carboxylic acid: $\text{HOOC}-\text{CH}_2-\text{COOH} \rightarrow \text{CH}_3\text{COOH} + \text{CO}_2$.	[1]
48	C	Option C is correct. Benzoic acid reacts with PCl_5 or SOCl_2 to replace the hydroxyl group with chlorine: $\text{C}_6\text{H}_5\text{COOH} + \text{PCl}_5 \rightarrow \text{C}_6\text{H}_5\text{COCl} + \text{POCl}_3 + \text{HCl}(\text{g})$, yielding the versatile synthetic intermediate benzoyl chloride.	[1]
49	B	Option B is correct. Reaction: $2\text{CH}_3\text{COOH}(\text{aq}) + \text{CaCO}_3(\text{s}) \rightarrow (\text{CH}_3\text{COO})_2\text{Ca}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$. The calcium carbonate marble chips dissolve slowly with steady bubbling of CO_2 gas.	[1]
50	A	Option A is correct. In $\text{HOOC}-\text{COOH}$, the second $-\text{COOH}$ group is directly bonded to the first. Because the carbonyl and hydroxyl groups are strongly electronegative, they exert a powerful $-I$ inductive withdrawal that stabilizes the $-\text{OOC}-\text{COOH}$ mono-anion, making K_a over 1000 times larger than that of ethanoic acid.	[1]
51	D	Option D is correct. Soap molecules are amphiphatic: the long non-polar hydrocarbon tail is lipophilic (hydrophobic) and embeds into oil/grease, while the ionic carboxylate head ($-\text{COO}^- \text{Na}^+$) is hydrophilic and interacts with water. Agitation breaks grease into tiny spherical droplets (micelles) surrounded by negative charges that repel each other, keeping grease stably suspended in wash water.	[1]
52	C	Option C is correct. Carboxylic acids react with basic ammonia in an acid-base neutralization: $\text{CH}_3\text{COOH} + \text{NH}_3 \rightarrow \text{CH}_3\text{COONH}_4$ (ammonium ethanoate). It is an ionic salt that crystallizes as white solid crystals.	[1]
53	A	Option A is correct. Heating ammonium carboxylate salts causes loss of a water molecule (pyrolysis/dehydration): $\text{CH}_3\text{COO}-\text{NH}_4^+ \xrightarrow{\text{heat}} \text{CH}_3\text{CONH}_2$ (ethanamide) + $\text{H}_2\text{O}(\text{g})$. This provides a classical synthetic pathway to primary amides.	[1]
54	C	Option C is correct. In CH_3COCl , the chlorine atom exerts a powerful electron-withdrawing inductive effect ($-I$) alongside the carbonyl $\text{C}=\text{O}$ group. This combined electron withdrawal weakens the adjacent C-H bonds and disperses negative charge in the conjugate carbanion/enolate.	[1]
55	B	Option B is correct. Carboxylic acids with two carboxyl groups on the same carbon (gem-dicarboxylic acids) undergo concerted decarboxylation through an intramolecular 6-membered cyclic transition state: a proton transfers to a carbonyl oxygen while CO_2 departs, requiring very low activation energy.	[1]
56	D	Option D is correct. Esterification is a slow, reversible condensation reaction: $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$. Concentrated sulfuric acid acts as a homogeneous acid catalyst (protonating the carbonyl oxygen to accelerate nucleophilic attack by ethanol) and absorbs water, shifting the position of equilibrium toward the ester product.	[1]
57	C	Option C is correct. The alcohol portion is attached to the oxygen: $-\text{CH}(\text{CH}_3)_2$ (an isopropyl or 1-methylethyl group). The acyl portion attached to the carbonyl is $\text{CH}_3\text{CH}_2\text{C}(=\text{O})-$ (a 3-carbon propanoate chain). Combining alcohol alkyl name first + acyl carboxylate suffix: 1-methylethyl propanoate (isopropyl propanoate).	[1]
58	A	Option A is correct. Acid hydrolysis is reversible: $\text{RCOOR}' + \text{H}_2\text{O} \rightleftharpoons \text{RCOOH} + \text{R}'\text{OH}$. In alkaline hydrolysis: $\text{RCOOR}' + \text{OH}^- \rightarrow \text{RCOO}^- + \text{R}'\text{OH}$. The strong base deprotonates the acid to form a carboxylate anion (RCOO^-). The negative charge on RCOO^- repels nucleophilic attack by the alcohol lone pair, preventing the reverse esterification reaction and driving the hydrolysis irreversibly to completion.	[1]
59	D	Option D is correct. Natural fats and oils are triglycerides (triesters of the trihydric alcohol glycerol with fatty acids). Alkaline hydrolysis (saponification) cleaves all three ester bonds: Triglyceride + $3\text{NaOH} \rightarrow$ Glycerol (propane-1,2,3-triol) + 3RCOONa (soap molecules with hydrophobic hydrocarbon tails and hydrophilic $-\text{COO}^- \text{Na}^+$ heads).	[1]
60	B	Option B is correct. Alkaline hydrolysis breaks the ester linkage into sodium benzoate ($\text{C}_6\text{H}_5\text{COONa}$) and methanol (CH_3OH). Subsequent addition of dilute HCl protonates the benzoate ion: $\text{C}_6\text{H}_5\text{COONa} + \text{HCl} \rightarrow \text{C}_6\text{H}_5\text{COOH}(\text{s}) + \text{NaCl}$, precipitating white crystals of benzoic acid.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
61	B	Option B is correct. Splitting the ester bond (-COO-): the acyl fragment is H-C(=O)- , derived from methanoic acid (HCOOH); the alkoxy fragment is $\text{-O-CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$, derived from butan-1-ol.	[1]
62	D	Option D is correct. Formula $\text{C}_3\text{H}_6\text{O}_2$ corresponds to two esters: (1) methyl ethanoate (acyl = ethanoate 2C, alkoxy = methyl 1C); (2) ethyl methanoate (acyl = methanoate 1C, alkoxy = ethyl 2C).	[1]
63	A	Option A is correct. The 4 isomeric esters of $\text{C}_4\text{H}_8\text{O}_2$ are: (1) methyl propanoate ($\text{CH}_3\text{CH}_2\text{COOCH}_3$); (2) ethyl ethanoate ($\text{CH}_3\text{COOCH}_2\text{CH}_3$); (3) propyl methanoate ($\text{HCOOCH}_2\text{CH}_2\text{CH}_3$); (4) 1-methylethyl methanoate ($\text{HCOOCH}(\text{CH}_3)_2$).	[1]
64	C	Option C is correct. Both have $M_r = 74.0$. Propanoic acid forms strong cyclic hydrogen-bonded dimers. Ethyl methanoate has no hydrogen atom bonded to oxygen; its intermolecular attractions are restricted to permanent dipole-dipole forces and London dispersion forces, requiring far less energy to vaporise.	[1]
65	A	Option A is correct. Although ester molecules cannot form hydrogen bonds among themselves, their electronegative oxygen atoms carry lone pairs that form hydrogen bonds with the δ^+ hydrogen atoms of water molecules ($\text{C=O} \cdots \text{H-O-H}$), allowing moderate solubility for small esters.	[1]
66	D	Option D is correct. Unlike aqueous equilibria where water is the bulk solvent and $[\text{H}_2\text{O}]$ is constant, in esterification water is formed as a chemical product in a non-aqueous organic mixture, and its concentration varies significantly. Hence, $[\text{H}_2\text{O}]$ must be explicitly included in the K_c expression.	[1]
67	B	Option B is correct. At equilibrium: $[\text{ester}] = 2/3$, $[\text{water}] = 2/3$, $[\text{acid}] = 1/3$, $[\text{alcohol}] = 1/3$. $K_c = (2/3 \times 2/3) / (1/3 \times 1/3) = (4/9) / (1/9) = 4.0$ (dimensionless, units cancel).	[1]
68	C	Option C is correct. By Le Chatelier's principle, adding an excess of alcohol shifts the position of equilibrium to the right. Removing water as it forms also shifts equilibrium to the right, driving the conversion of acid close to 100%.	[1]
69	B	Option B is correct. Isotopic labeling ($\text{CH}_3\text{COOH} + \text{CH}_3\text{-}^{18}\text{OH} \rightarrow \text{CH}_3\text{CO-}^{18}\text{OCH}_3 + \text{H}_2\text{O}$) proves that the oxygen atom in the ester linkage originates from the alcohol. The -OH group that forms water comes from the carboxylic acid: acyl-oxygen cleavage occurs, NOT alkyl-oxygen cleavage.	[1]
70	A	Option A is correct. The acid catalyst (H^+ from H_2SO_4) protonates the carbonyl oxygen: $\text{R-C(=O)OH} + \text{H}^+ \rightleftharpoons \text{R-C(=O}^+\text{-H)OH}$. Resonance transfers the positive charge onto carbon: $\text{>C}^+\text{-OH}$, dramatically increasing its electrophilicity and allowing attack by the weak nucleophile alcohol (R'OH).	[1]
71	D	Option D is correct. The reaction mixture contains unreacted acid and catalyst. Shaking with aqueous Na_2CO_3 neutralizes the acids ($2\text{H}^+ + \text{CO}_3^{2-} \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}$), producing visible bubbling. The resulting sodium salts dissolve in the bottom aqueous layer, allowing the immiscible upper ester layer to be isolated cleanly in a separating funnel.	[1]
72	C	Option C is correct. Anhydrous MgSO_4 or CaCl_2 pellets absorb residual dissolved water from the organic ester layer without reacting chemically, clumping until the liquid becomes completely clear and dry.	[1]
73	A	Option A is correct. The crude liquid contains unreacted alcohol (ethanol bp 78°C) and trace impurities. Fractional distillation isolates the ester by collecting the distillate boiling sharply at its precise boiling point (77°C for ethyl ethanoate).	[1]
74	C	Option C is correct. 3-methylbutyl ethanoate ($\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$, isoamyl acetate) has a distinct sweet banana/pear aroma, widely used in food flavorings and confections.	[1]
75	B	Option B is correct. Ethyl butanoate (ethyl butyrate) has a sweet, fruity odor resembling fresh pineapples, commonly used as a flavoring in juices, candies, and perfumes.	[1]
76	D	Option D is correct. Octyl ethanoate is the principal ester component providing the fresh, sweet, citrus aroma of oranges.	[1]
77	C	Option C is correct. Hydroxy-carboxylic acids with 4 or 5 carbon atoms (γ - or δ -hydroxy acids) undergo spontaneous intramolecular esterification upon warming, eliminating water to form stable 5- or 6-membered cyclic esters known as lactones (e.g. γ -butyrolactone).	[1]
78	A	Option A is correct. PET is a condensation polyester formed from benzene-1,4-dicarboxylic acid and ethane-1,2-diol. Boiling with concentrated aqueous NaOH cleaves the ester linkages, regenerating sodium terephthalate and ethane-1,2-diol (used in chemical recycling of plastic bottles).	[1]
79	D	Option D is correct. Saponification value measures the moles of ester bonds per gram of fat. Since each triglyceride molecule consumes 3 KOH molecules, fats with shorter fatty acid chains have more molecules (and more ester bonds) per gram, yielding higher saponification values.	[1]
80	B	Option B is correct. Industrial transesterification reacts natural plant oil triglycerides with excess methanol at 60°C in the presence of $\sim 1\%$ KOH or NaOCH_3 catalyst: Triglyceride + $3\text{CH}_3\text{OH} \rightleftharpoons$ Glycerol + 3 Fatty Acid Methyl Esters (FAME, biodiesel).	[1]
81	B	Option B is correct. Triglycerides have high molecular weights ($\sim 900\text{ g/mol}$) and viscosities 10 to 20 times higher than diesel, leading to poor atomization, injector clogging, and engine deposits. Converting them into methyl esters ($M_r \sim 280$) lowers viscosity to match petroleum diesel specifications.	[1]
82	D	Option D is correct. Alkaline hydrolysis of phenyl ethanoate cleaves the ester: $\text{CH}_3\text{COOC}_6\text{H}_5 + 2\text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{C}_6\text{H}_5\text{ONa} + \text{H}_2\text{O}$. Because phenol is weakly acidic ($\text{p}K_a 10$), it reacts with excess NaOH to form the water-soluble salt sodium phenoxide.	[1]
83	A	Option A is correct. The reverse of esterification: water attacks the carbonyl carbon via an addition-elimination pathway. The $\text{-}^{18}\text{OH}$ group from labeled water is incorporated into the carboxylic acid (forming $\text{CH}_3\text{CO-}^{18}\text{OH}$), while the departing ethoxide leaving group takes the original ^{16}O to form ordinary ethanol ($\text{CH}_3\text{CH}_2\text{-}^{16}\text{OH}$).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
84	C	Option C is correct. Salicylic acid contains both a phenolic -OH and a -COOH group. The phenolic -OH group is esterified with ethanoic anhydride in the presence of concentrated H ₃ PO ₄ : 2-HO-C ₆ H ₄ -COOH + (CH ₃ CO) ₂ O → 2-CH ₃ COO-C ₆ H ₄ -COOH (aspirin) + CH ₃ COOH.	[1]
85	A	Option A is correct. The ester linkage in aspirin is hydrolyzed by intestinal esterase enzymes and alkaline bile: 2-CH ₃ COO-C ₆ H ₄ -COO- + OH- → 2-HO-C ₆ H ₄ -COO- (active anti-inflammatory salicylate) + CH ₃ COO- (ethanoate).	[1]
86	D	Option D is correct. Paracetamol (acetaminophen, N-(4-hydroxyphenyl)ethanamide) is an amide (formed between 4-aminophenol and ethanoic anhydride), whereas aspirin is an ester (formed between the phenolic -OH of salicylic acid and ethanoic anhydride). Amides are significantly more resistant to hydrolysis than esters.	[1]
87	B	Option B is correct. In an ester (R-C(=O)O-R'), resonance donation from the alkoxy oxygen [R-C(=O)O-R' ↔ R-C(O-)=O+-R'] partially satisfies the electron deficiency of the carbonyl carbon. This +M resonance effect reduces the partial positive charge on carbon, making it far less electrophilic than in aldehydes and ketones.	[1]
88	C	Option C is correct. LiAlH ₄ cleaves the ester bond and reduces the acyl portion to a primary alcohol: CH ₃ CH ₂ COOCH ₂ CH ₃ + 4[H] → CH ₃ CH ₂ CH ₂ OH (propan-1-ol) + CH ₃ CH ₂ OH (ethanol).	[1]
89	B	Option B is correct. Ethyl ethanoate is an effective, pleasant-smelling, low-toxicity solvent with a boiling point of 77 °C. It rapidly dissolves resinous binders, nitrocellulose lacquers, and adhesives, evaporating smoothly to leave uniform polymer films.	[1]
90	A	Option A is correct. Natural waxes are solid, water-repellent esters composed of very long-chain saturated fatty acids (C ₁₆ -C ₃₆) esterified with long-chain primary aliphatic alcohols (C ₁₆ -C ₃₆), such as myricyl palmitate (C ₁₅ H ₃₁ COOC ₃₀ H ₆₁) in beeswax.	[1]
91	D	Option D is correct. Terylene is formed by polyester condensation: n HOOC-C ₆ H ₄ -COOH + n HO-CH ₂ CH ₂ -OH → [-CO-C ₆ H ₄ -COO-CH ₂ CH ₂ -O-] _n + (2n-1) H ₂ O. Ester linkages connect the alternating monomers into strong, crease-resistant textile fibers.	[1]
92	C	Option C is correct. Alpha-cleavage of the weak C-O ester bond adjacent to the carbonyl group expels a neutral methoxy radical (.OCH ₃ , 31) and generates the resonance-stabilized acylium cation: [CH ₃ -C=O]+ ↔ [CH ₃ -C#O+], which has m/z = 43.0 and constitutes the dominant base peak.	[1]
93	A	Option A is correct. Methyl ethanoate has structure CH ₃ -CO-O-CH ₃ . The acetate methyl (CH ₃ -CO-) has no neighboring hydrogens and appears as a 3H singlet at delta ~2.0 ppm. The methoxy methyl (-O-CH ₃) also has no neighboring hydrogens and appears as a 3H singlet at delta ~3.7 ppm, giving two singlets with area ratio 1:1.	[1]
94	C	Option C is correct. The rate law is second-order: Rate = k[ester][OH-]. When [OH-] is in large excess, its concentration does not change measurably during the reaction, collapsing the rate law to pseudo-first-order: Rate = k'[ester], where k' = k[OH-].	[1]
95	B	Option B is correct. Saturated fatty acids have straight, flexible chains that pack efficiently into crystalline lattices with strong London dispersion forces (high melting points, solids). Cis-C=C double bonds in plant oils introduce rigid 30° bends ('kinks') that prevent tight packing, drastically weakening intermolecular forces (low melting points, liquids).	[1]
96	D	Option D is correct. Partial hydrogenation uses nickel catalyst at 150 °C. The reversible addition of hydrogen to cis double bonds allows the intermediate radical/alkene to rotate and reform the thermodynamically more stable trans configuration, creating trans fats that raise LDL ('bad') cholesterol.	[1]
97	C	Option C is correct. Nucleophilic acyl substitution by ammonia: CH ₃ COOCH ₂ CH ₃ + NH ₃ → CH ₃ CONH ₂ + CH ₃ CH ₂ OH. Ammonia displaces the weaker ethoxide leaving group to form the primary amide ethanamide and ethanol.	[1]
98	A	Option A is correct. Reaction with the first equivalent of R'MgX displaces the alkoxy group to form a ketone intermediate (R-CO-R'). The ketone is more reactive than the ester and immediately reacts with a second equivalent of R'MgX to yield a tertiary alcohol containing two identical R' groups upon hydrolysis.	[1]
99	D	Option D is correct. The broad O-H absorption band (2500-3300 cm ⁻¹) is diagnostic for carboxylic acids. Esters lack an O-H group entirely, displaying only narrow C-H stretches (~2950 cm ⁻¹) alongside the sharp carbonyl stretch (shifted to ~1735-1750 cm ⁻¹ , higher than carboxylic acids at 1710 cm ⁻¹).	[1]
100	B	Option B is correct. Methyl salicylate possesses an ester group and a phenolic -OH group. Saponification cleaves the ester to carboxylate (-COO- Na+) and methanol. Because the phenolic -OH is acidic, excess NaOH neutralizes it to phenoxide (-O- Na+), forming disodium salicylate: 2-HO-C ₆ H ₄ -COOCH ₃ + 2NaOH → 2-NaO-C ₆ H ₄ -COONa + CH ₃ OH + H ₂ O.	[1]
101	C	Option C is correct. Ethanoic acid (1) has no halogens and is the weakest (pK _a 4.76). Chloroethanoic acid (2, pK _a 2.86) has one chlorine withdrawing electrons (-I). Fluoroethanoic acid (4, pK _a 2.59) is stronger than chloroethanoic because fluorine is more electronegative than chlorine. Dichloroethanoic acid (3, pK _a 1.29) has TWO chlorine atoms withdrawing electron density simultaneously, dispersing the negative charge more effectively than a single fluorine, making it the strongest of the four: 1 < 2 < 4 < 3.	[1]
102	A	Option A is correct. Initial moles of NaOH = 0.100 dm ³ * 1.00 mol/dm ³ = 0.100 mol. Moles of HCl used = 0.0500 dm ³ * 1.00 mol/dm ³ = 0.050 mol of unreacted NaOH remaining. Moles of NaOH consumed by ester = 0.100 - 0.050 = 0.050 mol. Since 0.050 mol of ester consumed exactly 0.050 mol of NaOH, the mole ratio is 1 : 1, confirming exactly one ester group per molecule.	[1]
103	D	Option D is correct. Hydrolysis yields an alcohol (P) and a carboxylic acid (Q). Compound P turns dichromate green (it is a primary or secondary alcohol) and forms a yellow precipitate with alkaline iodine (it contains the CH ₃ -CH(OH)- group), uniquely identifying P as propan-2-ol (CH ₃ -CH(OH)-CH ₃ , 3 carbons). Compound Q effervesces with NaHCO ₃ (it is a carboxylic acid). Since the ester has 5 carbons total, Q must have 5 - 3 = 2 carbons (ethanoic acid, CH ₃ COOH). Thus, ester X is 1-methylethyl ethanoate, CH ₃ COOCH(CH ₃) ₂ .	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
104	B	Option B is correct. Effervescence with Na_2CO_3 identifies W as a carboxylic acid: propanoic acid ($\text{CH}_3\text{CH}_2\text{COOH}$). Isomer Z is an ester. Hydrolysis of Z followed by acidification yields an alcohol that gives a yellow precipitate with alkaline iodine. The only alcohol with ≤ 2 carbons that gives a positive iodoform test is ethanol ($\text{CH}_3\text{CH}_2\text{OH}$). However, if the alcohol were ethanol, the ester would be ethyl methanoate ($\text{HCOOCH}_2\text{CH}_3$), where ethanol (P) gives iodoform, OR if the acid is ethanoic acid, the ester is methyl ethanoate ($\text{CH}_3\text{COOCH}_3$), where the ethanoate salt gives iodoform with iodine. In fact, ethanol from ethyl methanoate gives yellow CHI_3 , identifying Z.	[1]
105	A	Option A is correct. The combustion equation for ethyl propanoate ($\text{C}_5\text{H}_{10}\text{O}_2$): $\text{C}_5\text{H}_{10}\text{O}_2 + 6.5 \text{ O}_2 \rightarrow 5 \text{ CO}_2 + 5 \text{ H}_2\text{O}$. Each mole of ester contains 5 carbon atoms and produces 5 moles of CO_2 . Therefore, 0.10 mol of ester produces: $0.10 \text{ mol} \times 5 = 0.50 \text{ mol}$ of CO_2 gas. At RTP ($24.0 \text{ dm}^3/\text{mol}$): Volume = $0.50 \text{ mol} \times 24.0 \text{ dm}^3/\text{mol} = 12.0 \text{ dm}^3$.	[1]
106	C	Option C is correct. The nucleophilic oxygen atom of ethanol attacks the strongly electron-deficient carbonyl carbon of ethanoyl chloride (nucleophilic addition), forming a tetrahedral intermediate. The chloride ion, which is an excellent leaving group, is expelled as the $\text{C}=\text{O}$ double bond reforms (elimination). Deprotonation yields the ester and HCl : an addition-elimination mechanism.	[1]
107	B	Option B is correct. Trichloroethanoic acid (pK_a 0.65) is strongest due to $-I$ charge dispersal from three chlorines. Ethanoic acid (pK_a 4.76) is a weak carboxylic acid with carboxylate resonance. Phenol (pK_a 10.0) is much weaker, stabilized only by phenoxide π delocalization into the benzene ring. Ethanol (pK_a 16.0) is the weakest, destabilized by the $+I$ effect of the ethyl group: Trichloroethanoic > Ethanoic > Phenol > Ethanol.	[1]
108	D	Option D is correct. Moles of NaOH used = $0.0400 \text{ dm}^3 \times 5.00 \text{ mol/dm}^3 = 0.200 \text{ mol}$. Ratio of NaOH to starting diol = $0.200 \text{ mol} / 0.10 \text{ mol} = 2.0$. This confirms that the oxidation product is a diprotic dicarboxylic acid (bearing two $-\text{COOH}$ groups). Only primary alcohol groups ($-\text{CH}_2\text{OH}$) can be oxidized to carboxylic acid groups ($-\text{COOH}$). Secondary alcohols oxidize to ketones (which do not react with NaOH), and tertiary alcohols do not oxidize. Hence, both $-\text{OH}$ groups in the diol must be primary.	[1]
109	A	Option A is correct. Esters of methanoic acid (formate esters, $\text{H}-\text{COO}-\text{R}$) retain the formyl (aldehyde-like) hydrogen bonded directly to the carbonyl group ($\text{H}-\text{C}(=\text{O})\text{O}-$). Because of this formyl grouping, methanoate esters can be oxidized to carbonate species, reducing Tollens' reagent to form a brilliant silver mirror. Esters of all other carboxylic acids (ethanoates, propanoates) lack this hydrogen and show no reaction with Tollens' reagent.	[1]
110	C	Option C is correct. Overall percentage yield of a multi-step synthesis is the product of the individual step fractional yields: Overall yield = $0.800 \times 0.750 = 0.600 = 60.0\%$. To produce 0.60 mol of final product at 60.0% overall yield: Moles of starting ethene required = (desired moles / overall yield) = $0.60 \text{ mol} / 0.600 = 1.00 \text{ mol}$ of ethene.	[1]